Numerical simulations of internal solitary waves in a pycnocline by an internal wave beam

Yasaman Yazdani
PHY479
Supervisor: Prof. Grisouard

April 20, 2026

1 Introduction

Internal waves, driven by buoyancy associated with density stratification and influenced by the Coriolis force due to Earth's rotation [1], propagate along vertical density gradients in the ocean interior and, when they break, generate turbulence that mixes water masses [2]. This mixing redistributes heat and climatically important tracers, contributes significantly to the global ocean circulation, and makes understanding their generation mechanisms essential [3].

Internal solitary waves (ISWs) are large-amplitude, nonlinear internal gravity waves that propagate as isolated packets with horizontal scales near 1 km and vertical displacements up to 100 m [4].

ISWs are observed in lakes, the oceans, and the atmosphere, yet their generation is often difficult to attribute [5]. Proposed mechanisms include stratified flow over topography [8, 9], subharmonic interactions, where an internal tide transfers energy to lower-frequency waves that can grow and evolve into strongly nonlinear motions [10], steepening of internal tides, where a long internal-tide wavefront sharpens and breaks into solitary-wave packets [13], gravitational collapse of mixed patches, where a localized mixed region restratifies and releases internal wave energy [11, 12], and the generation by internal-tide beams impinging on a pycnocline, the strongly stratified transition layer between the upper mixed layer and the lower stratified layer [17], which is the focus of this project.

New & Pingree (1990) proposed a mechanism for generating ISWs, where an internal wave beam impinging on the pycnocline transfers energy to modal internal tides, which in turn generate ISWs. Later, Grisouard et al. (2011) investigated this using a two-dimensional, non-hydrostatic Boussinesq model in an idealized, non-rotating configuration with the MIT General Circulation Model (MITgcm), a finite-volume numerical circulation model [7]. The efficiency of this beam–pycnocline transfer is expected to depend sensitively on pycnocline thickness, which sets the vertical length scale of the stratification transition and controls how strongly the beam interacts with the interface. Quantifying this dependence is therefore central to assessing when the beam–pycnocline pathway robustly produces pycnocline-trapped ISWs [6].

Since the launch of the Surface Water and Ocean Topography (SWOT) satellite in December 2022, nonlinear internal-wave signatures have been observed at unprecedented spatial resolution, renewing interest in their dynamics and energy pathways [4].

Despite the importance of the beam-pycnocline mechanism, it has not been systematically revisited using modern, GPU-accelerated models, particularly in the presence of rotation, limiting our ability to test the robustness of the original results and to explore how the mechanism depends on key parameters.

To address this gap, in this project I will reproduce and then extend the ISW-generation mechanism of Grisouard et al. using Oceananigans.jl, a fast finite-volume solver for the non-hydrostatic Boussinesq equations. By leveraging GPU computing, Oceananigans enables the high resolution and broad parameter-space exploration required to resolve the nonlinear wave dynamics of interest more efficiently than traditional solvers [14]. This project investigates whether an internal wave beam impinging on a pycnocline robustly generates trapped ISWs and examines how the inclusion of the Coriolis force affects the generation mechanism.

2 Method

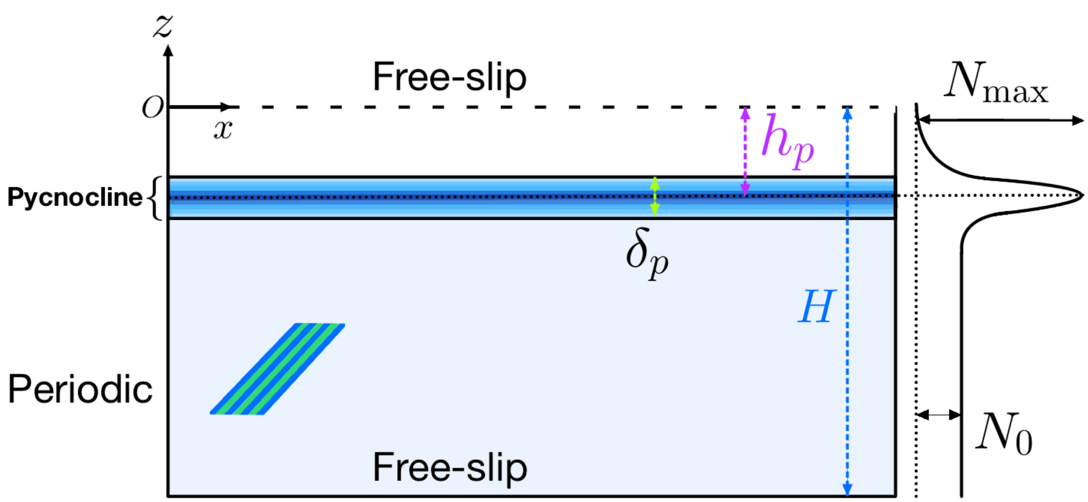


Figure 1: Numerical setup for internal wave beam–pycnocline interaction: a stratified 2D domain with periodic horizontal and free-slip vertical boundaries, where a Gaussian-forced internal wave beam impinges on a pycnocline of thickness δ_p centered at depth h_p, corresponding to the location of maximum stratification ($N_{\max}$), separating a uniform lower layer (N_0) from an upper mixed layer.

Table 1: Set-up parameters that vary from one experiment to another.

Designation of the experiment		**E1**	**E2**
Domain depth	H (m)	0.95	0.8
Domain length	L (m)	6	3
Brunt–Väisälä frequency in the lower layer	N_0	0.6 rad s^{-1}	0.6 rad s^{-1}
Forcing frequency	ω_0	0.4243 rad s^{-1}	0.424 rad s^{-1}
Vertical location of the centre of the pycnocline	h_p	2 cm	2 cm
Thickness of the pycnocline	δ_p	1 cm	1 cm
Kinematic viscosity	ν (m^2 s^{-1})	10^{-6}	10^{-6}
Mass diffusivity	κ (m^2 s^{-1})	1.43×10^{-7}	1.43×10^{-7}
Amplitude	A (cm)	1.4×10^{-2}	0.9×10^{-2}
Wavelengths of the forcing	λ_x, λ_z (cm)	53.6	24.7
Relative density jump	Δ_p (‰)	2.05	3.38
Duration of the experiments		$16T_0$	$12T_0$
Horizontal resolution	dx (mm)	4	2
Fine vertical resolution	dz_m (mm)	0.4	0.3
Coarse vertical resolution	dz_M (mm)	4	4
Sponge layer length	l_s (m)	1	0.1
Gaussian width in x	σ_x(m)	0.13	0.28
Gaussian width in z	σ_z(m)	0.13	0.28
Centre position in x	x_c(m)	0.6	0.6
Centre position in z	z_c(m)	-0.5	-0.8

The physical configuration of the setup is illustrated in Figure 1, with the specific values for all parameters discussed in this section detailed in Table 1.

Using Julia and Oceananigans v0.97.7, I implemented a two-dimensional, non-rotating, and non-hydrostatic internal-wave-beam setup that solves the incompressible Boussinesq equations. The model is defined on a Cartesian domain with $x \in [0, L_x]$ and $z \in [-H, 0]$, where L_x and H represent the horizontal and vertical extents of the domain, respectively.

The configuration employs periodic boundary conditions in the horizontal and free-slip, impermeable boundaries at the top and bottom. The model evolves the horizontal and vertical velocities, $u(x, z, t)$ and $w(x, z, t)$, along with the buoyancy tracer, $b(x, z, t)$, where t denotes time (s); it is initialized from rest and advanced with a time step of $\Delta t = 0.005$ s.

Following Grisouard et al. (2011), the stratification profile is continuous, consisting of a homogeneous upper layer and a stratified lower layer with constant buoyancy, separated by a pycnocline. The associated Brunt–Väisälä frequency profile, $N^2(z)$, is defined as:

$$N^2(z) = (g\Delta p)\,\frac{2}{\sqrt{\pi}\,\delta_p}\exp\left[-\left(\frac{2(z+h_p)}{\delta_p}\right)^2\right] + \begin{cases} N_0^2, & z < -h_p \\ 0, & z \geq -h_p \end{cases}$$

where g is the gravitational acceleration, while $\Delta\rho$, δ_p, and h_p represent the relative density change across the pycnocline, the pycnocline thickness, and the depth of its centerline, respectively. N_0 denotes the constant Brunt–Väisälä frequency of the lower layer.

The localized increase in stratification within the pycnocline extends the range of frequencies over which internal waves can be trapped, enabling the formation of waves that remain confined to this layer rather than propagating freely into the surrounding fluid.

To generate the internal wave beam, a localized, time-periodic body forcing is imposed via the terms $F_u(x,z,t)$, $F_w(x,z,t)$, and $F_b(x,z,t)$ in the horizontal momentum, vertical momentum, and buoyancy equations, respectively. To prevent unwanted reflections, a damping profile $\mu(x)$ is applied equally to both velocity components such that $\mu_u(x) = \mu_w(x) = \mu(x)$. The resulting governing equations for momentum and buoyancy are:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + w\frac{\partial u}{\partial z} - fv = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} + \nu\nabla^2 u + F_u - \mu_u(x)u$$

$$\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + w\frac{\partial w}{\partial z} = -\frac{1}{\rho_0}\frac{\partial p}{\partial z} + b + \nu\nabla^2 w + F_w - \mu_w(x)w$$

$$\frac{\partial b}{\partial t} + u\frac{\partial b}{\partial x} + w\frac{\partial b}{\partial z} = \kappa\nabla^2 b + F_b$$

where f is the Coriolis parameter and v is the transverse velocity component (both of which are set to zero for the primary configurations $E1$ and $E2$ in Table 1); ν is the kinematic viscosity; and κ is the mass diffusivity. In contrast to Grisouard et al. (2011), smaller diffusivities are replaced here with more realistic oceanic values for ν and κ. The body forcing terms are defined as:

$$F_u(x,z,t) = A\left[-\tilde{F}\frac{m}{k}\cos\phi - \frac{1}{k}\tilde{F}_z\sin\phi + \frac{m}{k^2}\tilde{F}_x\sin\phi\right]$$

$$F_w(x,z,t) = A\tilde{F}\cos\phi, \qquad F_b(x,z,t) = A\sqrt{N^2}\,\tilde{F}\sin\phi$$

In these expressions, A is the amplitude scale and $\tilde{F}(x,z) = \exp\left(-\frac{(x-x_c)^2}{2\sigma_x^2} - \frac{(z-z_c)^2}{2\sigma_z^2}\right)$ represents a Gaussian envelope centered at (x_c, z_c), where the widths σ_x and σ_z determine the horizontal and vertical decay scales. The phase is given by $\phi(x,z,t) = kx + mz - \omega_0 t$, where k and m are the horizontal and vertical wavenumbers, and ω_0 is the angular frequency chosen to satisfy the internal-wave dispersion relation. The forcing period is $T_0 = 2\pi/\omega_0$, and the spatial derivatives of the envelope are denoted by F_x and F_z. Finally, I set $m = -k$, corresponding to a 45° propagation angle.

The forcing terms added to the momentum equations are formulated following Zhou and Diamessis (2013) [16], while I modified the buoyancy forcing from their density forcing to match Oceananigans.jl's buoyancy-tracer formulation.

To minimize reflections, I added a boundary sponge layer of length ℓ_s starting at $x_{\text{sponge}} = L_x - \ell_s$. Within this region, the relaxation time $\tau(x)$ varies smoothly from $\tau_{\text{inner}} = T_0$ in the interior to $\tau_{\text{outer}} = T_0/1000$ at the boundary using a sigmoid function centered at $x_0 = x_{\text{sponge}} + \ell_s/2$, corresponding to the Herbaut formulation as described in Zhang & Marotzke (1999) [15]. This profile is defined as:

$$\mu(x) = \begin{cases} 0, & x \leq x_{\text{sponge}} \\ \dfrac{1}{\tau(x)}, & x > x_{\text{sponge}} \end{cases}, \quad \tau(x) = \tau_{\text{inner}}(1 - s(x)) + \tau_{\text{outer}}\, s(x), \quad s(x) = \frac{1}{1 + e^{-k_{\text{sig}}(x - x_0)}}$$

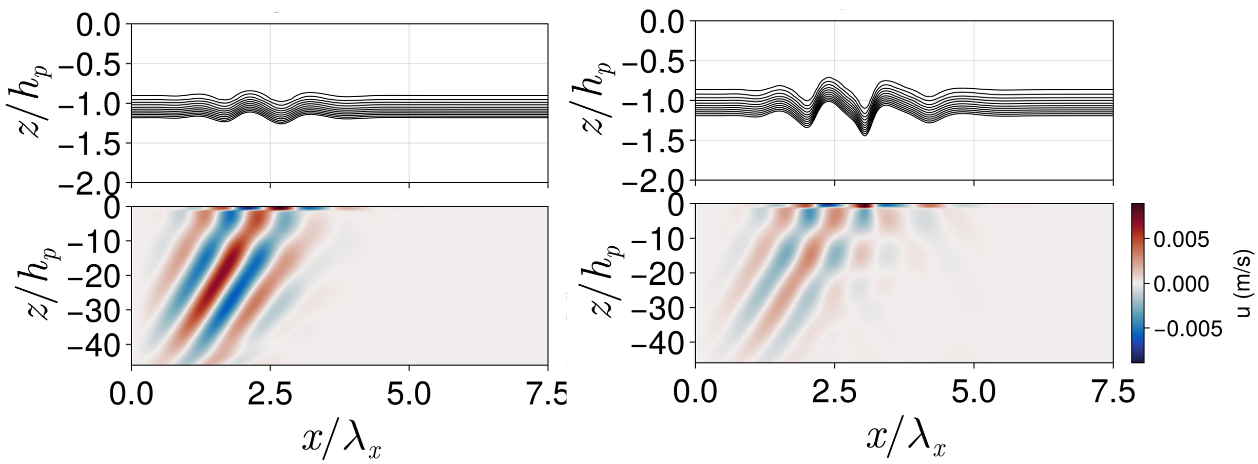
In the implementation, μ_u is sampled on x-faces for u and μ_w on x-centers for w to match the staggered grid.

I used a constant horizontal resolution Δx and a vertical resolution that varies smoothly from a coarse resolution Δz_M in the lower part to a fine resolution Δz_m in the upper part of the domain, with the middle of the transition zone set at $z = -3h_p$, and one grid cell in the y direction. This choice concentrates resolution near the pycnocline, where sharp gradients and nonlinear steepening occur, while reducing computational cost in less dynamically active regions.

To investigate the impact of rotation, two additional cases, $E1'$ and $E1''$, were constructed using the $E1$ parameter set with Coriolis frequencies of $f = N_0/3$ and $f = 2N_0/3$, respectively. These Coriolis parameters are significantly higher than those typically found in terrestrial geophysical contexts; these large values are selected primarily to illustrate the general influence of rotation rather than to represent specific realistic scenarios. In these rotating cases, all parameters remain identical to $E1$ except for the forcing frequency ω_0, which was adjusted to 0.4472 rad s^{-1} and 0.5099 rad s^{-1}, respectively. This adjustment ensures that the dispersion relation is satisfied while maintaining a constant propagation angle of 45°. Consequently, the momentum equation for the transverse velocity component v is given by:

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + w\frac{\partial v}{\partial z} + f\,u = \nu\nabla^2 v$$

3 Results and discussion



(a)

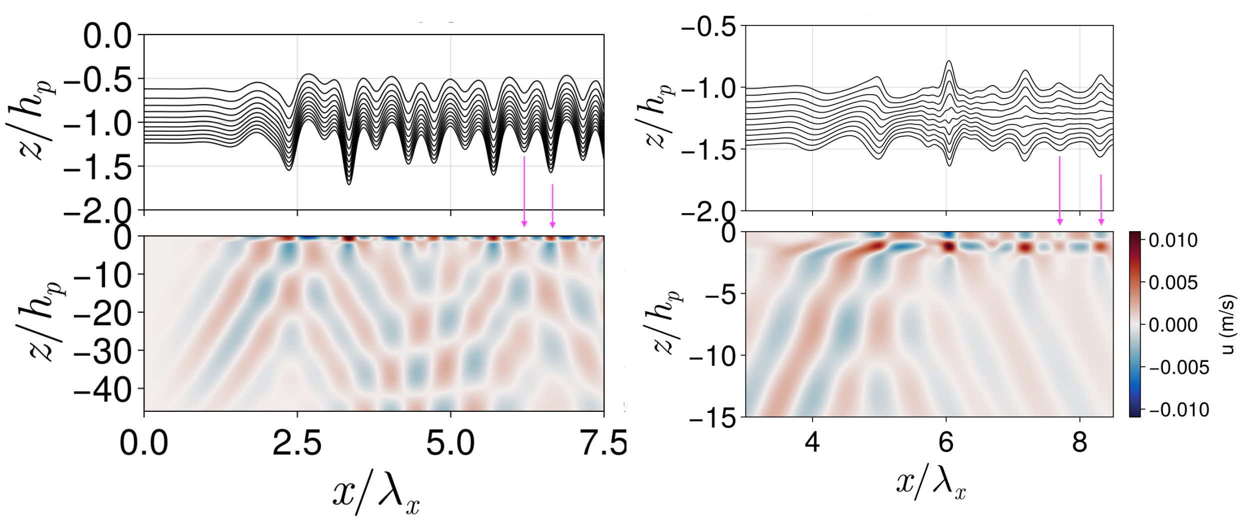
(b)

Figure 2: Interaction of the internal wave beam with the pycnocline in experiment E1. Top panels: Magnification of the isopycnals centred at $z = -h_p$. Bottom panels: Horizontal velocity field illustrating the spatial extent of the internal wave beam. (a) Initial interaction as the beam reaches the pycnocline, generating small-amplitude, nearly sinusoidal isopycnal displacements. (b) Subsequent evolution showing nonlinear steepening in the top panel and the reflected component of the wave beam in the bottom panel.

Results from experiment E1 are presented in Figures 2a,2b, and 3a. The top panels illustrate the contours of selected isopycnals within the pycnocline, while the bottom panels depict the spatial distribution of the horizontal velocity field across the full water depth. The color scale represents the horizontal velocity amplitude, where blue and red indicate negative and positive values, respectively.

Upon interaction with the pycnocline (Figure 2a), the internal wave beam undergoes partial transmission and reflection. The transmission process excites a wave through an essentially linear mechanism, as established by Gerkema (2001) and Akylas et al. (2007). This transmitted component is refracted due to the intensified local Brunt–Väisälä frequency within the pycnocline; it subsequently reflects off the upper boundary and is partially retransmitted into the lower layer, emerging at a greater horizontal distance than the initial reflection from the pycnocline base (Mathur & Peacock, 2009). As the experiment progresses (Figure 2b), nonlinear steepening becomes evident. This effect is eventually countered by dispersion, resulting in the formation of a mode-1 ISW, as shown in Figure 3a.

A similar evolution occurs in experiment E2, resulting in the generation of a mode-2 wave, as illustrated in Figure 3b. In both Subfigures 3a and 3b, the wave propagates to the right; the leading pulse exhibits a larger amplitude and is followed by a secondary pulse of smaller amplitude, as indicated by the pink arrows. The corresponding perturbations in the horizontal velocity field are evident in the lower panels of both figures. Specifically, the larger amplitude in Figure 3a (lower subpanel) correlates with a more intense horizontal red region. In Figure 3b, this manifests as a vertical structure consisting of an upper horizontal blue region and a lower horizontal red region, characteristic of a mode-2 structure. Furthermore, the color intensity in the leading column (rightmost) is more pronounced than in the subsequent column.



(a) (b)

Figure 3: Developed ISW structures in experiments E1 and E2. Top panels: Vertical isopycnal displacements showing the modal structure. Bottom panels: Corresponding horizontal velocity fields (u). (a) Generation of a mode-1 ISW in experiment E1, characterized by a single vertical maximum in displacement. (b) Generation of a mode-2 ISW in experiment E2, indicated by the characteristic vertical dipole in the velocity field (upper blue and lower red regions). In both cases, pink arrows highlight the leading larger-amplitude pulse followed by a smaller-amplitude secondary pulse, moving to the right.

Figure 4 depicts the spatiotemporal evolution of the vertical displacement (η) of the isopycnal at the center of the pycnocline (resting depth $z = -h_p$) for experiments $E1, E1'$, and $E1''$. The vertical displacement is normalized by the pycnocline depth, h_p. In all panels, the internal wave beam interacts with the pycnocline from the left. The response on the right side appears more dispersed, reflecting the increased propagation distance from the generation region. In the middle panel ($E1'$), the introduction of the Coriolis parameter ($f = N_0/3$) enhances dispersion, thereby disrupting the balance between nonlinear steepening and dispersive effects. Under stronger rotation ($f = 2N_0/3$, bottom panel), the initially sharp wave structure can no longer be sustained and instead evolves into a wider, more diffuse disturbance.

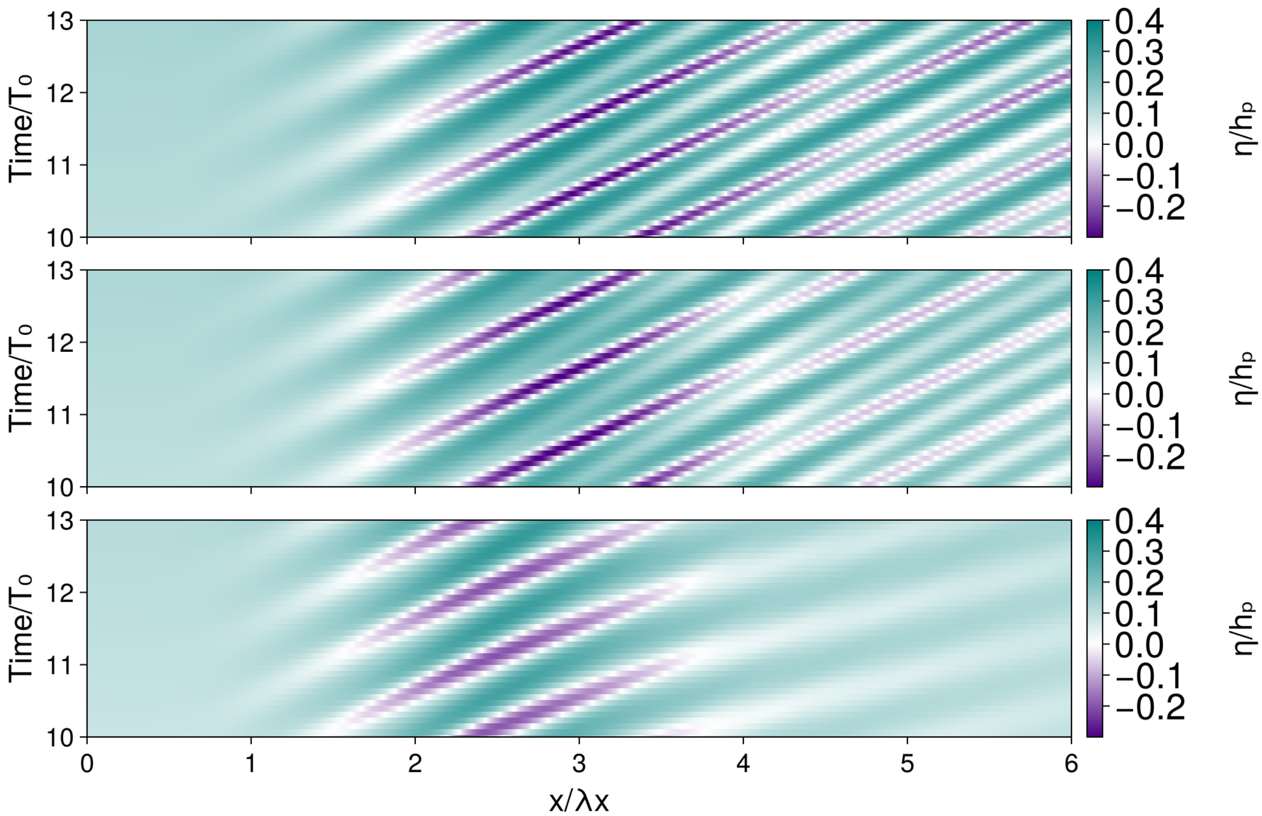


Figure 4: Spatiotemporal evolution of normalized isopycnal displacement at the pycnocline center ($z = -h_p$). Top: Non-rotating case (E1), showing a sharp, coherent structure. Middle: Moderate rotation (E1', $f = N_0/3$), where increased dispersion begins to disrupt the balance with nonlinear steepening. Bottom: Strong rotation (E1", $f = 2N_0/3$), where dominant rotational dispersion further disrupts this balance, resulting in a wider, more diffuse disturbance.

3.1 Mode Selection

Grisouard et al. (2011) demonstrated that the excitation of mode-n ISWs occurs when the horizontal phase speed of the generating internal wave beam falls within the range of phase speeds for mode-n waves trapped in the pycnocline. Their approach utilized linear theory to approximate modal properties, providing a framework to interpret subsequent nonlinear results. Following this methodology, we assume a vertical velocity solution to the linear, inviscid Boussinesq equations of the form:

$$w(x, z, t) = W(z) \exp(i(Kx - \Omega t))$$

Given our focus on rightward-propagating waves, we set $K > 0$. It follows that W and K must satisfy the eigenvalue problem:

$$\frac{d^2W}{dz^2} + K^2 \frac{N^2(z) - \Omega^2}{\Omega^2} W = 0$$

subject to the boundary conditions $W(0) = W(-H) = 0$.This system yields an infinite sequence of eigenfunctions, W_n, with corresponding eigenvalues, K_n, where n denotes the mode number. The general solution for the rightward-propagating vertical velocity field is expressed as:

$$w(x, z, t) = \sum_{n=1}^{\infty} a_n W_n(z) \exp(i(K_n x - \Omega t))$$

where a_n represents the modal amplitudes. Solving this system requires the buoyancy frequency profile $N(z)$ and the frequency Ω as inputs to determine W_n and K_n. The phase speed for each mode is then defined as $c_n = \Omega / K_n$.

Figures 4a and 4b display the computed phase speeds for the first three modes (c_1, c_2, c_3) for experiments $E1$ and $E2$, respectively, as a function of Ω.

As shown in Figure 5a, the curves for all three modes intersect the horizontal line at $c_n/v_\phi = 1$. However, since ISWs must be trapped within the pycnocline, we consider only solutions where $\Omega > N_0$. For experiment $E1$, this condition is satisfied only for mode 1, which explains the observed formation of mode-1 ISWs in this setup. Applying the same logic to the parameters of $E2$ (Figure 5b), the curve for mode 2 intersects the $c_n/v_\phi = 1$ threshold at a frequency where $\Omega > N_0$. While the mode-1 curve also eventually reaches this threshold, the resulting frequency exceeds the maximum value of $N(z)$ for this configuration. Consequently, mode 2 is the preferred selection observed in experiment $E2$.

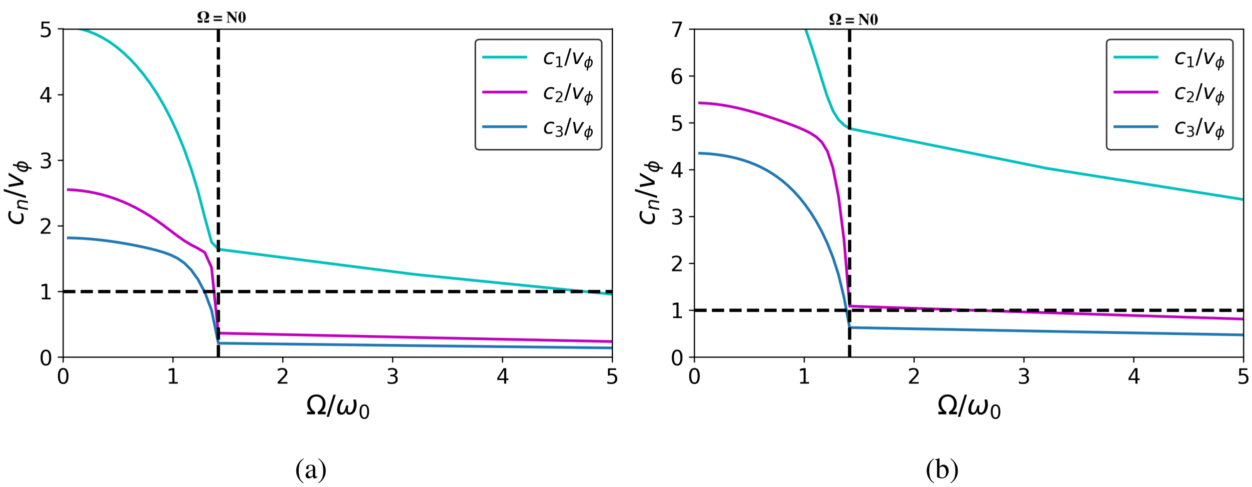


Figure 5: Phase speeds of the first three vertical modes normalized by the beam phase speed, shown as a function of normalized frequency for (a) experiment E1 and (b) experiment E2. The horizontal dashed line indicates the phase-speed matching condition, while the vertical dashed line marks the lower-layer stratification limit. Mode selection occurs where a mode curve intersects the phase-speed condition within the admissible frequency range.

4 Conclusion

This study shows that internal wave beam interaction with a pycnocline is a robust mechanism for ISW generation, with mode selection governed by phase-speed matching between the beam and pycnocline-trapped modes. The results also demonstrate that the Coriolis force disrupts the generation process: increasing the Coriolis parameter f enhances rotational dispersion, which weakens the balance with nonlinear steepening and inhibits the formation of sharp solitary waves, leading instead to broader, more diffuse disturbances.

Future work can evaluate the impact of pycnocline thickness on ISW amplitude to identify the threshold at which the beam-pycnocline pathway becomes energetically dominant. In addition, quantifying inter-modal energy transfer through temporal Fourier analysis would provide further insight into how energy is distributed among vertical modes.

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